

Hallucination Detection Report

Generated: 2026-04-15 14:26 · Extracted triples: 5 · Verdicts: 5 · Inconsistencies: 0 · Symbolic violations: 0

Executive Summary

2 Correct	2 Predicate Mismatch	1 Fabricated Fact
---------------------------------	--	---

Extracted Triples

All verifiable triples extracted from the LLM response before any canonicalization or checking.

Subject	Predicate	Object	Verifiable	Verification Type	Predicate Definition
400	HAS_DOWN_PAYMENT	19.4%	Yes	ENTITY_ATTRIBUTE	Indicates that the entity has a down payment percentage of 19.4%.
500	HAS_DOWN_PAYMENT	15.4%	Yes	ENTITY_ATTRIBUTE	Indicates that the entity has a down payment percentage of 15.4%.
430	HAS_ESTIMATED_DOWN_PAYMENT_RANGE	Down payment is between 19.4% for 400 and 15.4% for 500	Yes	FACTUAL_CLAIM	Indicates that the entity's estimated down payment falls within a specified range based on surrounding credit scores.
435	HAS_ESTIMATED_DOWN_PAYMENT_RANGE	Down payment is between 19.4% for 400 and 15.4% for 500	Yes	FACTUAL_CLAIM	Indicates that the entity's estimated down payment falls within a specified range based on surrounding credit scores.
433	HAS_ESTIMATED_DOWN_PAYMENT_RANGE	Down payment is between 19.4% for 400 and 15.4% for 500	Yes	FACTUAL_CLAIM	Indicates that the entity's estimated down payment falls within a specified range based on surrounding credit scores.

Canonicalized Triples

Triples after predicate normalization against the database schema. Shows the original predicate, the matched canonical predicate, and match confidence.

Subject	Original Pred.	Object	Canonical Pred.	Confidence
400	HAS_DOWN_PAYMENT	19.4%	DOWN_PAYMENT_REQUIRED_AS_UPFRONT_BORROWER_CASH_CONTRIBUTION	HIGH
500	HAS_DOWN_PAYMENT	15.4%	DOWN_PAYMENT_REQUIRED_AS_UPFRONT_BORROWER_CASH_CONTRIBUTION	HIGH
430	HAS_ESTIMATED_DOWN_PAYMENT_RANGE	Down payment is between 19.4% for 400 and 15.4% for 500	DOWN_PAYMENT_REQUIRED_AS_UPFRONT_BORROWER_CASH_CONTRIBUTION	MEDIUM
435	HAS_ESTIMATED_DOWN_PAYMENT_RANGE	Down payment is between 19.4% for 400 and 15.4% for 500	DOWN_PAYMENT_REQUIRED_AS_UPFRONT_BORROWER_CASH_CONTRIBUTION	MEDIUM
433	HAS_ESTIMATED_DOWN_PAYMENT_RANGE	Down payment is between 19.4% for 400 and 15.4% for 500	DOWN_PAYMENT_REQUIRED_AS_UPFRONT_BORROWER_CASH_CONTRIBUTION	MEDIUM

Logical Inconsistency Report

Contradictions detected between triples in the LLM response — independent of the database. These are self-contradictions within the answer itself.

No logical inconsistencies detected.

Symbolic Rule Violations

Violations caught by explicit domain constraints before database lookup. Violated triples were removed from the evidence retrieval step.

No symbolic rule violations detected.

Detected Hallucinations

Subject	Predicate	Object	Verdict	Match	Violated	Conf.	Explanation / Evidence
400	DOWN_PAYMENT_REQUIRED_AS_UPFRONT_BORROWER_CASH_CONTRIBUTION	19.4%	CORRECT	SEMANTIC		██████ ████ 0.80	The triple matches the DB candidate exactly in subject and predicate; the object '19.4%' semantically matches '19.4'. Evidence: subject: 400, predicate: DOWN_PAYMENT_REQUIRED_AS_UPFRONT_BORROWER_CASH_CONTRIBUTION, object: 19.4
500	DOWN_PAYMENT_REQUIRED_AS_UPFRONT_BORROWER_CASH_CONTRIBUTION	15.4%	CORRECT	SEMANTIC		██████ ████ 0.80	Subject and predicate match exactly; object '15.4%' is semantically equivalent to '15.4'. Evidence: subject: 500, predicate: DOWN_PAYMENT_REQUIRED_AS_UPFRONT_BORROWER_CASH_CONTRIBUTION, object: 15.4
430	DOWN_PAYMENT_REQUIRED_AS_UPFRONT_BORROWER_CASH_CONTRIBUTION	Down payment is between 19.4% for 400 and 15.4% for 500	PREDICATE_MISMATCH	NONE	PREDICATE	██████ ████ 0.60	The subject exists, but the predicate 'DOWN_PAYMENT_REQUIRED_AS_UPFRONT_BORROWER_CASH_CONTRIBUTION' is not a valid relation for '430' in DB. Evidence: subject: 430, predicate: RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL, object: deep subprime; subject: 430, predicate: INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE, object: 16.54
435	DOWN_PAYMENT_REQUIRED_AS_UPFRONT_BORROWER_CASH_CONTRIBUTION	Down payment is between 19.4% for 400 and 15.4% for 500 (estimated range)	PREDICATE_MISMATCH	NONE	PREDICATE	██████ ████ 0.60	The subject is found but does not have a valid DB entry for the predicate 'DOWN_PAYMENT_REQUIRED_AS_UPFRONT_BORROWER_CASH_CONTRIBUTION'. Evidence: subject: 435, predicate: RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL, object: deep subprime; subject: 435, predicate: INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE, object: 16.77
433	DOWN_PAYMENT_REQUIRED_AS_UPFRONT_BORROWER_CASH_CONTRIBUTION	Down payment is between 19.4% for 400 and 15.4% for 500 (estimated range)	FABRICATED_FACT	NONE	SUBJECT	██████ ████ 1.00	Neither subject nor object appears in any triple for this predicate in the DB; the fact is unsupported.

Triple Audit Trail (Original → Canonical → Verdict)

Full pipeline trace for every triple. UNVERIFIABLE* means the triple had no verdict (e.g. dropped due to output truncation).

Orig. Subject	Orig. Predicate	Orig. Object	Confidence	Canon. Predicate	Canon. Object	Verdict
400	HAS_DOWN_PAYMENT	19.4%	HIGH	DOWN_PAYMENT_REQUIRED_AS_UPFRONT_BORROWER_CASH_CONTRIBUTION	19.4%	CORRECT
500	HAS_DOWN_PAYMENT	15.4%	HIGH	DOWN_PAYMENT_REQUIRED_AS_UPFRONT_BORROWER_CASH_CONTRIBUTION	15.4%	CORRECT
430	HAS_ESTIMATED_DOWN_PAYMENT_RANGE	Down payment is between 19.4% for 400 and 15.4% for 500	MEDIUM	DOWN_PAYMENT_REQUIRED_AS_UPFRONT_BORROWER_CASH_CONTRIBUTION	Down payment is between 19.4% for 400 and 15.4% for 500	PREDICATE_MISMATCH
435	HAS_ESTIMATED_DOWN_PAYMENT_RANGE	Down payment is between 19.4% for 400 and 15.4% for 500	MEDIUM	DOWN_PAYMENT_REQUIRED_AS_UPFRONT_BORROWER_CASH_CONTRIBUTION	Down payment is between 19.4% for 400 and 15.4% for 500 (estimated range)	PREDICATE_MISMATCH
433	HAS_ESTIMATED_DOWN_PAYMENT_RANGE	Down payment is between 19.4% for 400 and 15.4% for 500	MEDIUM	DOWN_PAYMENT_REQUIRED_AS_UPFRONT_BORROWER_CASH_CONTRIBUTION	Down payment is between 19.4% for 400 and 15.4% for 500 (estimated range)	FABRICATED_FACT

Verdict Legend

Verdict	Description
CORRECT	All three components (subject, predicate, object) verified against DB.
RULE_VIOLATION	One or more components contradict DB evidence.
MULTI_RULE_VIOLATION	Violation spans multiple DB triples simultaneously.
UNVERIFIABLE	No usable DB evidence found for any component.
PREDICATE_MISMATCH	Subject real, but stated predicate is not a valid relation for it.
FABRICATED_FACT	Triple asserts a relationship that does not exist anywhere in DB.